Efficient Attention Mechanisms for Long-Context Summarization

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Abstract

We propose a novel sparse attention mechanism, called Chunked-Local Attention (CLA), designed to reduce the quadratic complexity of standard self-attention in transformer models when applied to long-document summarization tasks. Our method partitions the input sequence into overlapping chunks and applies local attention within each chunk, followed by a lightweight global aggregation step. We evaluate CLA on three long-document summarization benchmarks (arXiv, PubMed, and GovReport) and show that it achieves comparable ROUGE scores to full self-attention while reducing memory usage by 47% and inference time by 32%. We further conduct ablation studies to isolate the contribution of the global aggregation step.

1. Introduction

Transformer-based models have become the dominant architecture for text summarization. However, the quadratic complexity of self-attention with respect to sequence length limits their applicability to long documents such as scientific papers, legal contracts, and government reports, which often exceed 10,000 tokens.

Prior work has explored several directions to address this limitation, including sparse attention patterns (Longformer, BigBird), low-rank approximations (Linformer), and hierarchical encoding schemes. While these approaches reduce computational cost, they often trade off summarization quality, particularly for content that requires long-range reasoning across distant parts of the document.

In this paper, we introduce Chunked-Local Attention (CLA), a simple yet effective mechanism that balances efficiency and quality. Our key contributions are threefold:

(1) We propose the CLA mechanism combining local chunk attention with a lightweight global aggregation layer.

(2) We show CLA reduces memory usage by 47% and inference time by 32% compared to full self-attention, with minimal quality loss.

(3) We provide ablation studies analyzing the contribution of each component of CLA.

2. Related Work

Sparse Attention. Longformer and BigBird introduce fixed sparse attention patterns combining local windows with global tokens. Our work differs by using overlapping chunks with a learned aggregation step rather than fixed global tokens.

Efficient Transformers. Linformer and Performer approximate the attention matrix using low-rank projections or kernel methods. These methods scale linearly but can degrade performance on tasks requiring precise long-range dependencies, which we address by keeping exact local attention within chunks.

Long-Document Summarization. Prior summarization-specific models such as PEGASUS-X extend positional encodings for longer inputs but retain full attention complexity, limiting scalability.

3. Methodology

3.1 Chunked-Local Attention

Given an input sequence of length N, we partition it into overlapping chunks of size C with overlap O. Each chunk attends only to tokens within itself, reducing the attention complexity from $O(N^2)$ to $O(N * C)$.

3.2 Global Aggregation Layer

After local attention, we introduce a lightweight aggregation layer that summarizes each chunk into a fixed number of global tokens (K=8), which then attend to all other chunks' global tokens. This allows information to propagate across the full document with minimal additional cost.

3.3 Training Setup

We initialize our model from a pretrained BART-large checkpoint and fine-tune on each dataset for 10 epochs using AdamW with learning rate 3e-5, batch size 8, and chunk size C=512 with overlap O=64.

4. Experimental Setup

4.1 Datasets

We evaluate on three long-document summarization benchmarks:

- arXiv: scientific paper abstracts (avg. 6,900 tokens input)
- PubMed: biomedical article abstracts (avg. 4,700 tokens input)
- GovReport: US government reports (avg. 9,400 tokens input)

4.2 Baselines

We compare CLA against full self-attention Transformer, Longformer, BigBird, and Linformer, all fine-tuned under identical conditions.

4.3 Metrics

We report ROUGE-1, ROUGE-2, and ROUGE-L F1 scores, along with peak GPU memory usage (GB) and average inference time per document (ms).

5. Results

Table 1 summarizes our main results. CLA achieves ROUGE-L scores within 0.4 points of full self-attention on all three datasets, while using 47% less peak memory and reducing inference time by 32%.

Compared to Longformer, CLA achieves higher ROUGE-1 scores on arXiv (+1.2) and GovReport (+0.8), while using comparable memory. Compared to Linformer, CLA shows substantially better ROUGE-L scores (+2.1 on average), confirming that low-rank approximations lose important long-range information that CLA's global aggregation layer is able to preserve.

5.1 Ablation Studies

We conduct ablations removing the global aggregation layer entirely (Local-Only) and varying the number of global tokens K.

Removing global aggregation (Local-Only) causes a significant drop in ROUGE-L (-3.4 points on average), confirming its importance for capturing long-range dependencies across chunks.

Increasing K from 4 to 8 improves ROUGE-L by 0.9 points, but further increasing to K=16 yields diminishing returns (+0.1) while increasing memory usage by 12%, suggesting K=8 is a good efficiency/quality trade-off.

6. Discussion

Our results suggest that combining exact local attention with a lightweight global aggregation mechanism can recover much of the quality lost by purely sparse or low-rank approximations, while retaining most of their efficiency benefits.

We note that CLA's performance gains are most pronounced on documents exceeding 5,000 tokens; for shorter documents, the overhead of the aggregation layer slightly underperforms full attention.

Appendix A

Additional experimental details, hyperparameter search ranges, and per-dataset ROUGE breakdowns are provided in this appendix for reproducibility purposes.

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Appendix C

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7. Limitations

Our experiments are limited to English-language datasets; the behavior of CLA on morphologically richer or lower-resource languages remains untested. Additionally, CLA introduces two new hyperparameters (chunk size and overlap) that require tuning per task, unlike full self-attention which has none.

8. Conclusion

We presented Chunked-Local Attention (CLA), an efficient attention mechanism for long-document summarization that reduces memory usage by 47% and inference time by 32% compared to full self-attention, with minimal loss in summarization quality. Future work includes extending CLA to multilingual settings and exploring learned, adaptive chunk sizes.